ZLOG CLASS OF 2027
MZ 505
Inventory Systems

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Office Hours: By appointment

Course Description

Inventory Systems explores the essential concepts, models, and strategies that underpin effective inventory management in modern supply chains. The course equips students with the analytical tools and managerial insights needed to balance cost, service, and risk in inventory-related decisions.

Students will develop a solid foundation in both the theory and practice of inventory management, learning how to design, analyze, and improve inventory policies using quantitative models. Real-world case studies and a competitive supply chain simulation bring the concepts to life, illustrating how leading companies address inventory challenges in diverse industries. By the end of the course, students will be able to translate data into actionable decisions, critically evaluate inventory policies, and adapt strategies to dynamic business environments.

Requirements

The course assumes basic understanding of probability, statistics, algebra, calculus, and MS Excel. Students are advised to review these requirements before the start of the course and consult the instructor in case of doubts.

Attendance

It is assumed that students will attend all sessions. When absence is unavoidable, both the instructor and the program director should be notified in advance.

Evaluation

The final grade is made of four parts: homework (35%), class participation (15%), midterm exams (40%), and supply chain game & report (10%).

Homework assignments: There will be **regular** and **quiz** problems in the homework assignments. For regular problems it is allowed to talk to classmates, but each student should write and submit an individual work and acknowledge explicitly who are the students with whom she/he collaborated. For quiz problems no collaboration is allowed and the students should only communicate with the instructor regarding questions (this should be viewed as a take-home exam).

Mid-term Exams: There will be two midterm exams, each counting 20% of your final grade. The exams will be held as indicated in the course schedule. They will be closed book, but one A4 size double-sided "cheat sheet" is allowed. You will also need to bring a calculator.

Supply chain game: The students will be assigned into teams and will work together in a supply chain simulation game. Each team will have to prepare its strategy for the game, play the game and then prepare a report. The entire work on the game should be done exclusively within the team.

Class Participation: Students are expected to prepare all the pre-assigned readings and be active and effective participants during the class. Class participation will be determined on the basis of your comments and questions in each class session. Particularly during case studies, it is essential that you come to class prepared to share your insights and analysis with others, including being ready to open the case assigned for that day. Students will be evaluated on the quality (not the quantity) of the contributions. The criteria used to judge effective participation includes:

1. Is the participant a good listener and open to evaluating opposing perspectives in a constructive manner?
2. Are the points made relevant to the current discussion? Are they linked to the comments of others?
3. Do the comments demonstrate appropriate and insightful analysis of case data? Do they present a clear statement of the selected concepts and issues?
4. Does the participant integrate concepts from prior classes and assigned readings into current discussion where possible?

Reference Textbooks

1. Sunil Chopra and Peter Meindl, "Supply Chain Management: Strategy, Planning, and Operation" (5th Edition), 2011
2. Gerard Cachon and Christian Terwiesch, "Matching supply with demand - an introduction to operations management" (3rd Edition), 2012

Schedule[1]

Session	Topic	Date	Time	Details
01	Introduction	Sep 7th	08:30-10:00	Course Introduction
02	Economic Order Quantity 1	Sep 9th	08:30-10:00	The basic EOQ model
03	Economic Order Quantity 2	Sep 10th	08:30-10:15	EOQ Model extensions
04	TA session	Sep 15th	08:30-10:30	EOQ
05	Case Study	Sep 17th	08:30-10:00	Case study: Three Jays Corporation
06	Newsvendor Model 1	Sep 21st	08:30-10:00	The basic Newsvendor Model
07	Newsvendor Model 2	Sep 23rd	08:30-10:00	NV Model / Performance measures
08	Newsvendor Model 3	Sep 25th	08:30-10:00	NV application in contracting and revenue management
09	TA session	Sep 28th	08:30-10:30	NV Model
10	Case Study	Sep 30th	08:30-10:30	Case study: Sports Obermeyer, Ltd.
11	Order-up-to Model	Oct 2nd	08:30-10:00	Multi-period inventory policies; Order-up-to model
12	Order-up-to Model	Oct 5th	08:30-10:00	Order-up-to policy / Performance measures
13	Review Session	Oct 7th	08:30-10:00	Midterm review session
14	TA session	Oct 13th	08:30-10:30	Midterm preparation
15	**Midterm Exam**	Oct 19th	09:00-11:30	
16	Simulation Game	Oct 21st	08:30-10:30	Supply chain simulation game
17	(r, Q) Model	Oct 26th	08:30-10:15	Continuous review inventory policies; (r, Q) model
18	Risk Pooling 1	Oct 28th	08:30-10:00	Risk pooling strategies
19	Risk Pooling 2	Nov 4th	08:30-10:00	Risk pooling strategies
20	Case Study	Nov 6th	08:30-10:15	Case study: Four Star industries Singapore
21	Demand Forecasting 1	Nov 9th	08:30-10:00	Forecasting models
22	Demand Forecasting 2	Nov 11th	08:30-10:00	Forecasting models
23	Case study	Nov 16th	08:30-10:00	Case study: Wilkins, A Zurn Company – Demand Forecasting
24	TA Session	Nov 18th	08:30-10:30	Mid-term exam preparation
25	**Midterm Exam**	Nov 23rd	09:00-11:30	
26	Aggregate Production Planning	Nov 25th	08:30-10:00	Case study: Wilkins, A Zurn Company – Aggregate Production Planning
27	Simulation game	Dec 1st	11:30-13:30	Instructions for playing the game; One round of the short version of the 3-day simulation game

[1] Please consult the ZLOG calendar for any changes in the schedule

28	**Simulation Game**	Dec 2-5th	TBD	The supply chain game (3-day simulation)
29	Segmentation	Dec 9th	08:30-10:00	Approaches to SC segmentation
30	Wrap Up	Dec 10th	08:30-10:30	Recap of the semester and team presentations of the game results

Case Questions and Supporting Material

Session 5: "Three Jays Corporation" (HBS 9-915-531)

To prepare before the session: Case study questions (you can also download the case exhibits in excel format through the HBP course-pack)

1. Using the data in case Exhibit 4 and the 2012 annual demand, calculate the EOQ and ROP quantities for the five SKUs scheduled to be produced in the last week of June. How do these amounts compare with those calculated in 2011? Compare the increases in EOQs with the increases in annual demand.
2. Brodie is uncertain if the costs presented in case Exhibit 2 are appropriate for determining the EOQs. What changes would you recommend, and why? Should the cost of the three idle part-time workers be included when the production line is down? Using the 2012 annual demand, and your recommendation, recalculate the EOQs for the five SKUs.
3. Compare your results with those obtained using the data in case Exhibit 2. What do you attribute the differences to? After speaking to Jake and Josh, Brodie is now not sure if the EOQ model is the most appropriate for the current production process. Evaluate the scheduling method that Jake and Josh are using. Why are they not following the established system?
4. Compare the established EOQ/ROP procedure (described in case exhibit 2) with the one that Jake and Josh are using. Which system do you prefer? What improvements do you recommend?
5. What recommendations should Brodie present to Jana Fremont at his next meeting with her?

Session 10: "Sport Obermeyer Ltd." (HBS 9-695-022)

To prepare before the session: Case study questions (you can also download the case exhibits in excel format through the HBP course-pack)

1. Using the sample data given in Exhibit 10, make a recommendation for how many units of each style Wally Obermeyer should order during the initial phase of production. Assume that all ten styles in the sample problem are made in Hong Kong (at this point ignore the minimum production constraints), and that Obermeyer's initial production commitment must be at least 10,000 units. (Ignore price differences among styles in your initial analysis)
2. Now consider the same problem but with the minimum order quantity constraints (i.e., minimum order quantity of 600 units per style, provided any quantity of a style is ordered).

Propose a heuristic to solve that problem. That is, identify the major considerations that affect this decision, and then propose a reasonable heuristic that takes into account at least some of these considerations.

3. What operational changes would you recommend to Wally to improve performance?
4. How should Obermeyer management think (both short-term and long-term) about sourcing in Hong Kong versus China?

Session 20: "Four Star Industries Singapore - Matching Supply with Demand" (ABCC-2009-005)

To prepare before the session: Case study questions

1. What are the problems faced by Four Star Industries in their mattress manufacturing operations?
2. How does the proliferation of mattress varieties impact Four Star's operations?
3. Analyze the various courses of action that Sia Meng is contemplating to solve Four Star's operational problems. How effective will these actions be?
4. Discuss possible ways in which Four Star can reconfigure its product design, manufacturing and order fulfilment. How will this help to solve its operational problems? Please use qualitative and quantitative analysis to support your recommendations.

Session 23: "Wilkins, A Zurn Company: Demand Forecasting" (Ivey 906D06)

To prepare before the session: Case study questions (you can also download the case exhibits in excel format through the HBP course-pack)

1. How accurate was the demand forecast for the first quarter of 2005? Please use mean error (ME) and mean absolute per cent error (MAPE) to measure bias and magnitude, respectively.
2. What is the current demand forecasting method? Who uses the demand forecast? What are the consequences, if any, if the forecast is inaccurate?
3. Create a demand forecast for the PVB product family for the next three quarters. How does your demand forecast compare to Wilkins' demand forecast?
4. As Bernie Barge, what would you recommend to management and why? How should Barge convince management to follow his recommendations? Develop an action plan.

Session 26: "Wilkins, A Zurn Company: Aggregate Production Planning" (Ivey 906D17)

To prepare before the session: Case study questions

1. What is the goal of aggregate production planning (APP)?
2. How might APP benefit Wilkins? What are the risks of using APP at Wilkins?
3. What constraints are the most critical in the development of APP at Wilkins? Why?
4. Calculate a suggested safety inventory level for PVBs for each quarter using the formula:

$$SS = z^*\sigma_t^*sqrt(L) \quad (\text{where } \sigma_{Q1} = 1{,}877;\ \sigma_{Q2} = 3{,}361;\ \sigma_{Q3} = 3{,}851;\ \sigma_{Q4} = 8{,}154)$$

5. Assuming that overtime is not an option, create up to three APPs for the PVB product family:
    a. One utilizing a chase strategy,
    b. One utilizing a level strategy, and
    c. One utilizing a mixed optimal strategy, if the chase and level strategies are not optimal
5. Interpret your results. What are the benefits and limitations of each plan?
6. As Connors, would you choose to implement an APP? If so, which strategy would you suggest and why? If you would choose not to implement an APP, are there any additional changes that could be made to reduce inventory and improve inventory control at Wilkins?

To answer the case questions, please consider the following:

APP purpose, definitions, and strategies: https://en.wikipedia.org/wiki/Aggregate_planning

You can assume that the lead time is equal to 2 weeks and that plant operates 5 days/week, 13 weeks/quarter.

Formulate your APP in excel and (if possible) use solver to come up with the optimal plans per strategy. An outline of the variables to be used follows:

- Target: minimize total cost of production
- Changing cells: production per quarter (in units)
- Parameters: production cost (/unit), holding cost (/unit/quarter), hiring cost, safety stock (/quarter), initial inventory, initial number of employees, forecast sales, production rate (units/employee/quarter)
- Dependent variables: inventory holding cost, regular production cost, number of employees, employee change, employee cost
- Constraints: these depend on the strategy (level, chase, or mixed)

Conduct

Professional conduct is built upon the idea of mutual respect. Such conduct entails (but is not necessarily limited to):

- Arriving on time. Late arrivals are disruptive to both lectures and class discussion, and show disrespect to those who are on time.
- Minimizing disruptions. Cell phones should be turned off during class. You should avoid engaging in side conversations after class has begun.
- Focusing on the class. While you may take notes on laptops, do not use laptop computers or hand-held devices for other tasks while in class. Activities such as net surfing and answering email are very impolite and disruptive both to neighbors and the class.
- Respect. You should act respectfully toward all class participants.